FuelPrint Field plan

Aditya Singh's part · shared so everyone knows what Field and the SMS alerts do · version 2, 18 Oct 2026

What changed in version 2

- **Autopilot.** The approved mission flies by itself if no visitor takes it within 60 seconds. A visitor can still take over.
- **Automatic capture.** At the target, sampling starts by itself. The QC person still confirms each gas.
- **SMS alerts** are part of Aditya Singh's area: the `/alerts/` sign-up page and the alerts package.
- **Builders:** Kiro writes all Field and alerts code; Claude designs the 3D look.

What Field is

Field is the 3D world where the approved measurement actually happens. A drone leaves the depot, flies to the approved point, and samples the air at the real position of its inlet. What it reads depends on where the inlet really was and for how long, so a lazy flight gives a worse sample. Visitors can fly it from their own phone; if nobody does, the autopilot flies it.

Field is built on Aditya Singh's Mac. Kiro writes the code (server package `fuelprint.field` and `apps/field`). Claude designs the look (scene, lighting, materials, the mission screen, the phone page). Nobody else edits `apps/field` or `fuelprint.field`.

What is in the 3D world

Thing	Where it comes from
Real context buildings	The Kundli map crop in `data/context/derived-scene-crop.json` (991 footprints). At most 300 are drawn as boxes, all at an assumed 8 m height. Credited to Overture Maps, Google Open Buildings, Microsoft and OpenStreetMap.
Six fictional factories S01 to S06	`world-profile.json`: 80 × 60 m sheds with 8 m roofs. Stacks drawn up to their release height (25 to 45 m), labelled "synthetic".
Four sensor boxes N1 to N4	Poles with the inlet at 3 m, a status light and a small reading label
Roads	The rectangle from −400 to 400 m east and −350 to 250 m north, a centre cross and a spur, 6 m wide
Depot	At the origin. The drone starts and returns here.
Wind	An arrow and drifting dust specks that show direction and speed only
The drone	Body, rotors, and a separate inlet marker so everyone can see where the air is taken. A small tag shows who is flying: visitor, operator or autopilot.
Reading trail	Coloured dots along the drone path showing what it actually measured
Alert areas	The four fictional areas from `alert-areas.json` as faint labelled ground outlines; an area glows when it was sent a smoke warning

There is **no smoke plume coming out of the real source**. Drawing it would give away the hidden answer. Only measured values appear in the world. In the finale, a model prediction overlay can be added, clearly labelled as an estimate.

The drone, in numbers

Rule	Value
Top speed	3 m/s, acceleration 1 m/s²
Allowed area	East and north within ±450 m, height 2 to 80 m
Clearance	2 m from any building or stack; the server refuses to fly through them
Battery	64 Wh usable, 12 Wh reserve
Power use	Hover 160 W + instruments 10 W + speed² W
Server tick	20 times a second; screens get 10 updates a second

The server moves the drone, not the browser. The phone (or the autopilot) only says "I want to go this way, this fast". If the return trip would eat into the reserve, the server sends the drone home by itself (`return_to_depot`, reason BATTERY_RESERVE).

A mission, step by step

1. Operations approves a mission (it was proposed automatically). It appears on laptop 2 with its target point, height band and method.
2. The field operator presses "Let a visitor fly" and a 6-letter code plus QR appears for 120 seconds.
3. A visitor opens `/field/controller` on their phone (same Wi-Fi), types the code, sees a two-word phrase, and the operator confirms the phrase matches. The visitor joins the queue.
4. The first visitor in the queue presses "Fly" and holds the lease. One pilot at a time.
5. **If nobody takes the lease within 60 seconds, the autopilot starts by itself** and flies the approved route. The screen counts down before it starts. A visitor can take over at any time except during collection.
6. They (or the autopilot) fly to the target. Readings at the inlet change as they move.
7. At the target, **the capture starts by itself** and runs in stages: setup 30 s, background 30 s, purge 30 s, settle at least 120 s, collect 30 valid seconds, recover at least 120 s, then review 30 s. While the drone hovers still, these stages can run 15 times faster so a visitor does not wait 6 minutes. The pilot must hold position; the autopilot does this exactly.
8. The drone returns to the depot (by itself if the autopilot is flying).
9. Each gas has already been pre-checked automatically. The QC reviewer on laptop 2 confirms each channel. A channel that broke a rule cannot be passed.
10. Each channel becomes a `screening_record` (VALID, LIMITED or REJECTED with reasons), which Cases and Operations receive. The ranking updates by itself.

The capture rules

From the method `TEACH-NODE-ONLINE-2`:

- SO2 branch pulls 0.5 L/min, CO2 branch 0.3 L/min, each within ±5 %.
- Collection needs 30 consecutive valid seconds. A gap or fault ends the run of valid seconds; two good pieces are never glued together.
- 30 valid seconds gives 0.25 L of SO2 sample and 0.15 L of CO2 sample.
- Settling must reach a model error of 5 % or less, after at least 120 s.
- In the blocked inlet scenario, SO2 flow drops to zero at collection second 10. The SO2 number on screen still looks normal, but it is flagged FLOW_FAULT and adds no valid

volume. The 10 good seconds before the fault (0.083 L) are kept in the record but do not make a valid 30 s window, so SO2 is REJECTED. CO2 can still finish and be VALID.

Phone, keyboard and autopilot rules

Rule	Value
Input rate	About 10 messages a second
Dead man	No fresh input for 400 ms means the server stops the drone (target: stopped within 500 ms)
Lease	2 s, renewed only by fresh valid input
Leaving the page	Phone sends one "neutral" message; the drone stops
Reconnect	Stays stopped until the visitor presses "Fly" again; old input is never replayed
Take over	The operator can take control at any time; the visitor's or autopilot's lease ends first
Keyboard on laptop 2	W A S D to move, R and F for up and down, Space to stop
Autopilot start	By itself 60 s after approval with no pilot (if `auto_autopilot` is on), or by the operator or supervisor
Autopilot limits	Exactly the same speed, acceleration, area, clearance and battery rules as a person
Autopilot hand-over	A visitor can take over except during collection; when the visitor stops, the autopilot resumes after 10 s

The phone page is two large touch pads (left: move, right: up and down), a stop button, the lease timer, and the drone's live height and battery. It needs no camera, no location and no app install.

SMS alerts (also Aditya Singh's area)

- **Sign-up page `/alerts/`.** A visitor scans the alerts QR at the stall, types their mobile number, picks one or more of the four areas, ticks the consent line, and presses "Sign me up". The page shows the masked number and a "Remove me" button. Big buttons, no app install.
- **What they receive.** A normal SMS starting "FuelPrint DEMO (simulated):" when smoke heads to their area, and another when the board decides. No factory is ever named.
- **Who sends.** One teammate's Android phone with a SIM runs the free "SMS Gateway for Android" app on the stall Wi-Fi. The server tells it what to send. On every other laptop the server only writes the SMS to the screen (LOG_ONLY).
- **Limits.** One smoke warning per area every 10 minutes, at most 80 SMS a day. Anything over a limit is shown as "suppressed" with the reason.
- **Privacy.** Numbers are stored only in one table, never shown in full on any screen, never sent to the chat model, and deleted at midnight (or earlier with "Delete all phone numbers").

What Field gives others

To	What
Operations	Mission state changes (LEASED to COMPLETE) and who is flying
Cases and Operations	`screening_record` per channel after the review
Everyone	Vehicle positions in every tick
Operations (OP05, OP07)	`notification` and `alert_subscription` records (masked) from the alerts package

What Field needs

From	What
Core	Server, sessions, `AutomationBus`, `/ws/live`, `/ws/pilot`, records
Science	`ObservationProvider` readings at the drone inlet each tick; `CaptureRules` for settle error
Operations	Missions in state APPROVED; `investigation.warned_area_ids`; board decisions with `notice.area_ids`
A teammate	One Android phone with a SIM and an SMS pack as the SMS sender at the stall

Dates

Date	Field work
Mon 19 Oct	Field package skeleton, vehicle physics with limits and collision checks (Kiro)
Tue 20 Oct	Pairing, queue, lease, phone page, autopilot, capture stages, QC review screen, alerts package and `/alerts/` page (Kiro). 3D scene and drone look (Claude).
Wed 21 Oct	Join day; rehearsal on real phones (Android Chrome and iPhone Safari); real SMS test through the Android phone; frame rate check

The 3D view must hold at least 30 frames per second on laptop 2. If it can't, it drops detail, never science.

Finale work

- Instrument cutaway: the inside of the sampler with its two gas branches, pumps after the sensors, flow meters, and the separate particle path.
- 3D lab workbench showing the real specimen record, quantity, blank and QC results from Cases.
- The ground van on the road network, with its own inlet at 3 to 6 m.
- A model prediction overlay, labelled "estimate, not measured".
- A practice score based only on following the procedure (approval, readiness, proper rejection, safe return), never on guessing the factory.
- Reply STOP by SMS to unsubscribe.
- Reduced-motion mode and a 2D fallback map that uses the same server rules.